

The Finite-Dimensional Convex-Hull-Deviation Bound for ℓ_p^n , $p > 2$

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Status: candidate full solution, likely valid. Question 1 of Ivanov's arXiv:1501.02596 has an affirmative answer. The proof is a mixed-norm interpolation estimate for the centering operator. The separate exactness problem in Question 2 is not addressed.

1 Source question

For a Banach space X , its convex-hull-deviation constant is

$$\zeta_X = \sup_{D \subset B_X} \sup_{x \in \text{co } D} \inf_{y \in D} \|x - y\|_X.$$

Ivanov proves an upper bound for L_p spaces and, for $1 \leq p \leq 2$, the finite-dimensional refinement

$$\zeta_{\ell_p^n} \leq \left(2 \frac{n-1}{n}\right)^{|1/p-1/p'|}, \quad \frac{1}{p} + \frac{1}{p'} = 1. \quad (1)$$

Question 1 on arXiv PDF page 8 asks whether (1) also holds for $2 < p < \infty$ [1].

Remark 3. If $1 \leq p \leq 2$, then, using in the proof of Theorem 2 inequality (8) instead of (7), we arrive at:

$$(10) \quad \zeta_{\ell_p^n} \leq \left(2 \frac{n-1}{n}\right)^{\left|\frac{1}{p} - \frac{1}{p'}\right|}.$$

Still without any answer remains the following questions:

Question 1. Is the inequality (10) true if $p \in (2; \infty)$?

Question 2. Is the estimate in the inequality (2) exact in case of $p \in (1; \infty)$, $p \neq 2$?

The answer is affirmative. We use one earlier reduction from the source: if $\dim X = n$, then every value below ζ_X has a witness consisting of at most n points of B_X ; see Remark 2 on arXiv PDF page 3 [1]. The new ingredient is the weighted estimate below.

2 Idea of the proof

Fix weights α_i and consider the linear centering operator

$$T_\alpha(x_1, \dots, x_k) = \left(x_i - \sum_{j=1}^k \alpha_j x_j \right)_{i=1}^k.$$

At the Hilbert endpoint, the variance identity makes T_α a contraction from outer ℓ_∞ to weighted outer ℓ_2 . At the ℓ_∞ endpoint, expanding every centered vector as $\sum_j \alpha_j(x_i - x_j)$ bounds the weighted outer ℓ_1 norm by $2(1 - \sum_i \alpha_i^2)$. Mixed-norm Riesz–Thorin interpolation between these two estimates simultaneously changes the inner exponent from 2 to p and the outer exponent from 2 to p' . The interpolation parameter is $1 - 2/p$, exactly the exponent missing in the source question.

3 Weighted centering inequality

Lemma 3.1. *Let $2 < p < \infty$, let $p' = p/(p - 1)$, and let $\alpha_1, \dots, \alpha_k > 0$ sum to 1. For $x_1, \dots, x_k \in \ell_p^n$, put $x_0 = \sum_i \alpha_i x_i$. Then*

$$\left(\sum_{i=1}^k \alpha_i \|x_i - x_0\|_p^{p'} \right)^{1/p'} \leq \left(2 \left(1 - \sum_{i=1}^k \alpha_i^2 \right) \right)^{1-2/p} \max_i \|x_i\|_p. \quad (2)$$

Proof. Let $T_\alpha(x)_i = x_i - x_0$ and set

$$C_\alpha = 2 \left(1 - \sum_i \alpha_i^2 \right).$$

On ℓ_2^n , the Hilbert-space variance identity gives

$$\begin{aligned} \sum_i \alpha_i \|T_\alpha(x)_i\|_2^2 &= \sum_i \alpha_i \|x_i\|_2^2 - \left\| \sum_i \alpha_i x_i \right\|_2^2 \\ &\leq \left(\max_i \|x_i\|_2 \right)^2. \end{aligned}$$

Thus

$$T_\alpha : \ell_\infty^k(\ell_2^n) \longrightarrow \ell_2^k(\alpha; \ell_2^n) \quad \text{has norm at most 1.}$$

At the other endpoint,

$$\begin{aligned} \sum_i \alpha_i \|T_\alpha(x)_i\|_\infty &\leq \sum_{i,j} \alpha_i \alpha_j \|x_i - x_j\|_\infty \\ &\leq 2 \sum_{i \neq j} \alpha_i \alpha_j \max_j \|x_j\|_\infty \\ &= C_\alpha \max_j \|x_j\|_\infty. \end{aligned}$$

Consequently

$$T_\alpha : \ell_\infty^k(\ell_\infty^n) \longrightarrow \ell_1^k(\alpha; \ell_\infty^n) \quad \text{has norm at most } C_\alpha.$$

Apply the mixed-norm Riesz–Thorin theorem [2, Section 14] with

$$\theta = 1 - \frac{2}{p}.$$

The inner target and domain exponent satisfies $1/p = (1 - \theta)/2 + \theta/\infty$, while the outer target exponent satisfies

$$\frac{1}{p'} = \frac{1 - \theta}{2} + \frac{\theta}{1}.$$

The interpolated operator therefore obeys

$$\|T_\alpha x\|_{\ell_{p'}^k(\alpha; \ell_p^n)} \leq C_\alpha^\theta \|x\|_{\ell_\infty^k(\ell_p^n)},$$

which is exactly (2). □

4 Full answer to Question 1

Theorem 4.1. *For every integer $n \geq 2$ and every $2 < p < \infty$,*

$$\zeta_{\ell_p^n} \leq \left(2 \frac{n-1}{n} \right)^{1-2/p}$$

and hence inequality (10) in Ivanov’s paper holds throughout $2 < p < \infty$.

Proof. Use the finite-support reduction established in Remark 2 of the source [1]. It suffices to control a convex combination $x_0 = \sum_{i=1}^k \alpha_i x_i$ with $k \leq n$, positive weights summing to 1, and $x_i \in B_{\ell_p^n}$.

By Lemma 3.1,

$$\begin{aligned} \min_i \|x_i - x_0\|_p &\leq \left(\sum_i \alpha_i \|x_i - x_0\|_p^{p'} \right)^{1/p'} \\ &\leq \left(2 \left(1 - \sum_i \alpha_i^2 \right) \right)^{1-2/p}. \end{aligned}$$

Since Cauchy–Schwarz gives $\sum_i \alpha_i^2 \geq 1/k \geq 1/n$, the last expression is at most

$$\left(2 \left(1 - \frac{1}{n} \right) \right)^{1-2/p}.$$

Taking the supremum over the source’s finite witnesses proves the claim. Finally, for $p > 2$ one has $|1/p - 1/p'| = 1 - 2/p$, so the displayed estimate is precisely (1). $\square$

5 Verification, scope, and novelty

The proof uses only the source’s at-most- n -point reduction, the Hilbert variance identity, two triangle inequalities, Cauchy–Schwarz, and standard mixed-norm interpolation. The exponent bookkeeping is explicit in the proof. A deterministic randomized stress test is supplied in `code/check_mixed_norm_bound.py`; it is not part of the proof.

A bounded novelty search through 22 July 2026 checked the run registry and solution/attempt indexes, the local arXiv corpus, exact-title and keyword arXiv/web searches, and Ivanov’s later related arXiv:1904.06729. No explicit answer to Question 1 was found. Novelty confidence is moderate rather than high because the interpolation argument is short and may be implicit in earlier mixed-norm literature.

This packet fully answers Question 1 only. Question 2, concerning exactness of the dimension-free L_p estimate, remains outside the result. Human review should focus on the mixed-norm interpolation identity and the exact scope of the source’s finite-support reduction.

References

- [1] G. Ivanov, *Convex hull deviation and contractibility*, arXiv:1501.02596, 2015.
- [2] L. R. Williams and J. H. Wells, *Embeddings and Extensions in Analysis*, Springer-Verlag, 1975.